📌 SQL Aggregate Functions – Notes

🔷 1. Introduction

Aggregate Functions in SQL are used to perform calculations on multiple rows of data and return a **single summarized value**.

They help in:

- Analyzing trends
- Summarizing large datasets
- Performing statistical calculations
- Generating reports

🔷 2. Key Features

✓ Return a **single value**
✓ Work with numeric columns (mostly)
✓ Ignore **NULL values** (except COUNT(*))
✓ Often used with:

- GROUP BY
- HAVING
- ORDER BY

🔷 3. Basic Syntax

AGGREGATE_FUNCTION(column_name)

Example:

SELECT SUM(Salary) FROM Employee;

🔷 4. Common Aggregate Functions

1️⃣ COUNT()

Used to count rows in a table.

Types of COUNT:

✅ COUNT(*)

Counts all rows (including rows with NULL values).

SELECT COUNT(*) AS TotalRecords FROM Employee;

✓ Counts every row in the table.

✅ COUNT(column_name)

Counts only non-NULL values.

SELECT COUNT(Salary) AS NonNullSalaries FROM Employee;

✓ Ignores NULL salaries.

✅ COUNT(DISTINCT column_name)

Counts unique non-NULL values.

SELECT COUNT(DISTINCT Salary) AS UniqueSalaries FROM Employee;

✓ Ignores duplicates and NULL values.

2️⃣ SUM()

Used to calculate the total of a numeric column.

✅ SUM(column_name)

Adds all non-NULL values.

SELECT SUM(Salary) AS TotalSalary FROM Employee;

✓ Returns total salary
✓ Ignores NULL values

✅ SUM(DISTINCT column_name)

Adds only unique values.

SELECT SUM(DISTINCT Salary) AS DistinctSalarySum FROM Employee;

✓ Avoids duplicate salary values

3️⃣ AVG()

Used to calculate the average of a numeric column.

Formula:

AVG = SUM of non-NULL values / Count of non-NULL rows

✅ AVG(column_name)

SELECT AVG(Salary) AS AverageSalary FROM Employee;

✓ Calculates average of all non-NULL salaries.

✅ AVG(DISTINCT column_name)

SELECT AVG(DISTINCT Salary) AS DistinctAvgSalary FROM Employee;

✓ Average of unique salary values
✓ Ignores duplicates and NULLs

4️⃣ MIN() and MAX()

Used to find smallest and largest values.

✅ MAX()

SELECT MAX(Salary) AS HighestSalary FROM Employee;

✓ Returns highest non-NULL salary.

✅ MIN()

SELECT MIN(Salary) AS LowestSalary FROM Employee;

✓ Returns lowest non-NULL salary.

🔷 5. NULL Handling Summary

Function	Ignores NULL?
COUNT(*)	❌ No (counts all rows)
COUNT(column)	✅ Yes
SUM()	✅ Yes
AVG()	✅ Yes
MIN()	✅ Yes
MAX()	✅ Yes

🔷 6. Using with GROUP BY (Important for Interviews)

Aggregate functions are often used with GROUP BY.

Example:

```
SELECT Department, SUM(Salary)

FROM Employee

GROUP BY Department;
```

✓ Calculates total salary per department.

🔷 7. Practical Applications

✅ Payroll calculation
✅ Sales reporting
✅ Performance analysis
✅ Data summarization
✅ Business intelligence dashboards

🔷 8. Interview Important Points

- Aggregate functions return **one summarized value**
- They ignore NULL values (except COUNT(*))
- Used with GROUP BY and HAVING
- DISTINCT removes duplicates before calculation
- Very important in reporting and analytics roles

🎯 Conclusion

SQL Aggregate Functions:

- Help summarize large datasets
- Improve data analysis efficiency
- Essential for analysts, developers, and data professionals